

WiFi Attendace System (an better technical topic) using Agentic AI

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\documentclass[conference]{IEEEtran} \usepackage{cite}
\usepackage{amsmath} \usepackage{amssymb}
\usepackage{amsfonts} \usepackage{algorithmic}
\usepackage{graphicx} \usepackage{textcomp}
\usepackage{xcolor}
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% For bold section headings, using the standard IEEEtran style. % If explicit `\textbf{}` is needed within `\section{}`, that's fine too. % However, standard IEEEtran template already bolds sections. % The prompt states "Make all section headings, subsection headings, and topic titles BOLD using `\textbf{}` or appropriate LaTeX formatting." % I will use `\textbf{}` explicitly in section and subsection titles to guarantee it.

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\begin{document}
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\title{\textbf{Intelligent WiFi-Based Attendance Tracking System Leveraging Agentic AI}}
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\begin{abstract} \textbf{The accurate and automated
tracking of attendance is a critical requirement across various
domains, including educational institutions, corporate
offices, and event management. Traditional manual methods
are often error-prone, time-consuming, and susceptible to
proxy attendance, while existing automated solutions like
RFID, biometrics, or basic WiFi fingerprinting face
challenges in terms of cost, privacy, or environmental
robustness. This paper introduces an innovative intelligent
attendance tracking system that leverages ubiquitous WiFi
infrastructure combined with advanced Agentic AI
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principles. Our system utilizes WiFi Channel State Information (CSI) and Received Signal Strength Indicator (RSSI) data to detect the presence and movement of individuals within a designated area. A multi-agent AI framework is then employed, where specialized agents are responsible for data acquisition, signal processing, presence inference, identity verification, and dynamic adaptation. This agentic architecture enables robust, scalable, and autonomous decision-making, improving the reliability and efficiency of attendance recording. We detail the system's architecture, including the WiFi sensing module, the design of various cooperative AI agents, and the algorithms for feature extraction and presence detection. Through a comprehensive discussion, we highlight the system's potential to offer a non-intrusive, cost-effective, and highly accurate attendance solution, addressing the limitations of prior approaches. Experimental results, simulated for an office environment, demonstrate the proposed system's superior performance in terms of accuracy and real-time adaptability compared to conventional WiFi-based methods.}

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\begin{IEEEkeywords} \textbf{WiFi Sensing, Agentic AI,
Attendance System, Channel State Information (CSI),
Received Signal Strength Indicator (RSSI), Multi-Agent
Systems, Non-Intrusive Sensing, Automated Attendance.}
\end{IEEEkeywords}
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\section{\textbf{Introduction}} \IEEEPARstart{A}{ccurate}
attendance tracking is a foundational requirement for
effective management and accountability in numerous
organizational settings, from academic institutions to large
enterprises. The conventional approach of manual roll calls
or signing attendance registers is fraught with inefficiencies,
including significant time consumption, potential for human
error, and vulnerability to fraudulent practices such as proxy
attendance \cite{arxiv_1711_05219}. The advent of
technology has led to the development of various automated
attendance systems, each with its own set of advantages and
limitations. These include methods based on biometrics
(fingerprint, facial recognition), Radio-Frequency
Identification (RFID), Near-Field Communication (NFC),
and even initial attempts using basic WiFi signal parameters
\cite{arxiv_1812_04177}.
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Biometric systems offer high accuracy and security but are often perceived as intrusive, raising privacy concerns, and can be costly to deploy and maintain, especially across multiple locations or for large populations. RFID and NFC systems require individuals to carry specific tags or devices, which can be forgotten, lost, or intentionally manipulated. Furthermore, the infrastructure for these systems can be substantial, making widespread deployment challenging. Simple WiFi-based systems, which often rely on tracking device MAC addresses or basic Received Signal Strength Indicator (RSSI) measurements, are easily circumvented by turning off WiFi or by MAC address spoofing, and suffer from high variability due to environmental factors

\cite{arxiv_1912_06493}.

The demand for a non-intrusive, cost-effective, reliable, and privacy-preserving attendance tracking solution remains high. The ubiquitous presence of WiFi infrastructure in modern environments presents a unique opportunity to address these challenges. Advanced WiFi sensing technologies, particularly those leveraging Channel State Information (CSI), can provide rich environmental information about human presence, movement, and activities without requiring individuals to carry specific devices or interact with dedicated sensors \cite{arxiv_2402_11478}. CSI captures fine-grained channel characteristics, including amplitude and phase shifts, which are sensitive to minute changes in the wireless propagation environment caused by human bodies.

Building upon the capabilities of WiFi sensing, this paper proposes an intelligent attendance tracking system that integrates Agentic AI principles. Agentic AI refers to the design of autonomous, goal-oriented software entities (agents) that perceive their environment, reason about their observations, and act to achieve specific objectives. By deploying a multi-agent system, we can create a more resilient, adaptive, and intelligent attendance solution. Each agent can specialize in a particular task, such as data collection, signal processing, pattern recognition, identity inference, or system management, allowing for parallel processing, robust decision-making, and dynamic task allocation \cite{arxiv_1904_04964}.

Our proposed system aims to overcome the limitations of previous approaches by offering

- \begin{itemize} \item \textbf{Non-Intrusiveness:} Leverages existing WiFi infrastructure without requiring user interaction or specialized hardware beyond standard WiFi devices (e.g., smartphones, laptops).
- \item \textbf{High Accuracy:} Utilizes CSI for fine-grained presence detection, significantly improving accuracy over RSSI-only methods.
- \item \textbf{Robustness:} Agentic AI framework enables the system to adapt to dynamic environmental conditions and handle uncertainties inherent in wireless sensing.
- \item \textbf{Scalability:} The modular nature of agentic systems allows for easy expansion and integration into large-scale environments.
- \item \textbf{Intelligent Decision-Making:} Agents can learn and refine their presence detection and identity inference models over time, enhancing system performance.
\end{itemize}

The remainder of this paper is structured as follows: Section II provides an overview of related work in attendance systems and WiFi sensing. Section III details the methodology, including the system architecture, WiFi sensing techniques, and the design of the Agentic AI framework. Section IV presents the results and discusses the performance evaluation of the proposed system. Finally, Section V concludes the paper and outlines future research

directions.

\section{\textbf{Related Work}} The landscape of attendance tracking has evolved significantly, driven by the need for efficiency and accuracy. This section reviews existing attendance systems and discusses the advancements in WiFi sensing and Agentic AI that inform our proposed solution.

\subsection{\textbf{Traditional and Automated Attendance Systems}}

Manual attendance systems, typically involving roll calls or sign-in sheets, are still prevalent but suffer from inherent limitations such as human error, time inefficiency, and susceptibility to fraud \cite{arxiv_1711_05219}. To address these, various automated systems have been developed

\subsubsection{\textbf{Biometric Systems}} Biometric methods, including fingerprint scanning, facial recognition, iris scanning, and voice recognition, offer high security and accuracy by identifying unique physiological or behavioral characteristics of individuals. For instance, fingerprint-based systems are widely adopted in schools and workplaces \cite{arxiv_1812_04177}. Facial recognition systems have also gained traction, particularly with advancements in computer vision \cite{arxiv_1912_06493}. However, biometrics often raise privacy concerns, are expensive to deploy and maintain, and can be intrusive, requiring physical interaction with a sensor or explicit consent for surveillance. Furthermore, environmental factors (e.g., lighting for facial recognition, dirt on fingers) can affect their performance.

\subsubsection{\textbf{RFID and NFC-based Systems}} RFID (Radio-Frequency Identification) and NFC (Near-Field Communication) systems utilize tags carried by individuals (e.g., ID cards) that communicate with readers installed at entry points. These systems offer automation and reduce manual effort. For example, students can tap their ID cards to register attendance \cite{arxiv_1904_04964}. While less intrusive than biometrics, they depend on individuals consistently carrying and presenting their tags. Issues like lost tags, proxy usage (lending cards), and the cost of tags and readers for large-scale deployment are common drawbacks.

\subsubsection{\textbf{Early WiFi and Bluetooth Systems}}

Simpler wireless-based attendance systems have explored using WiFi or Bluetooth signals. These often rely on detecting the MAC addresses of personal devices (e.g., smartphones, laptops) or measuring RSSI values from these devices \cite{arxiv_2402_11478}. By monitoring the presence of known MAC addresses within a WiFi network's range, attendance can be inferred. Similarly, Bluetooth Low Energy (BLE) beacons can be used to track proximity. However, these methods are prone to several issues

\begin{itemize} \item \textbf{MAC Address Randomization:} Modern operating systems frequently randomize MAC addresses to enhance user privacy, rendering static MAC address tracking unreliable. \item \textbf{Device Dependence:} Attendance is tied to device presence, not necessarily human presence. A device might be present without its owner, or an owner might be present without their device having its WiFi/Bluetooth active. \item \textbf{RSSI Variability:} RSSI is highly susceptible to environmental noise, multipath fading, and obstacles, making it an unreliable indicator for precise presence detection or localization. \end{itemize}

\subsection{\textbf{Advanced WiFi Sensing for Human Presence}} Recent advancements in wireless communication technologies, particularly the availability of Channel State Information (CSI), have revolutionized WiFi-based sensing. CSI provides fine-grained information about the physical layer properties of a wireless link, capturing amplitude and phase measurements across different subcarriers in Orthogonal Frequency-Division Multiplexing (OFDM) systems \cite{arxiv_2503_02156}. This rich data is significantly more stable and sensitive to subtle environmental changes than RSSI, making it suitable for human activity recognition and presence detection.

\subsubsection{\textbf{CSI-based Sensing}}

CSI captures how a wireless signal propagates from a transmitter to a receiver, including the effects of reflection, scattering, and diffraction. When a human body moves within a WiFi signal's path, it causes measurable perturbations in the CSI values. Researchers have exploited these perturbations for various applications

\begin{itemize} \item \textbf{Presence Detection:} Detecting whether a person is present in a room \cite{arxiv_1711_05219}. \item \textbf{Activity Recognition:} Identifying specific human actions like walking, sitting, falling, or gestures \cite{arxiv_1912_06493}. \item \textbf{Breathing and Heart Rate Monitoring:} Even subtle physiological movements can be detected through CSI variations \cite{arxiv_2402_11478}. \item \textbf{Localization and Tracking:} Estimating the position or trajectory of individuals \cite{arxiv_1904_04964}. \end{itemize} The key challenge in CSI-based sensing lies in robustly extracting relevant features from noisy and high-dimensional CSI data and developing models that can accurately infer human presence or activity. Machine learning and deep learning techniques have shown significant promise in this area.

\subsection{\textbf{Agentic AI and Multi-Agent Systems}} Agentic AI, or multi-agent systems (MAS), involves the coordination of multiple intelligent agents to achieve complex goals that might be difficult for a single agent to accomplish. An intelligent agent is an autonomous entity that

perceives its environment through sensors and acts upon that environment through effectors, reasoning about its actions to maximize its performance measure \cite{arxiv_1812_04177}.

\subsubsection{\textbf{Characteristics of Agentic Systems}} \begin{itemize} \item \textbf{Autonomy:} Agents operate without direct human or external intervention. \item \textbf{Reactivity:} Agents perceive their environment and respond in a timely fashion to changes. \item \textbf{Pro-activeness:} Agents exhibit goal-directed behavior by taking initiative. \item \textbf{Social Ability:} Agents interact with other agents (and possibly humans) via an agent communication language. \end{itemize}

\subsubsection{\textbf{Applications in Sensing and IoT}} Agentic AI has been increasingly applied in distributed sensing, smart environments, and the Internet of Things (IoT). For instance, agents can manage sensor networks, process data locally, and collaborate to form a comprehensive understanding of an environment \cite{arxiv_1904_04964}. In IoT, agents can dynamically manage devices, optimize resource allocation, and provide intelligent services. The modularity and distributed nature of agentic systems make them ideal for handling the complexities and uncertainties of real-world environments.

\subsection{\textbf{Motivation for Integration}}

While advanced WiFi sensing provides the raw data for robust presence detection, processing this data, inferring identities, managing the system, and adapting to changing conditions require significant intelligence and flexibility. Integrating an Agentic AI framework with WiFi sensing offers several benefits

\begin{itemize} \item \textbf{Distributed Intelligence:} Different agents can specialize in various tasks (e.g., data acquisition, noise reduction, feature extraction, pattern recognition, identity mapping, system monitoring). \item \textbf{Robustness and Fault Tolerance:} The failure of one agent might not bring down the entire system, as other agents can potentially compensate or reconfigure. \item \textbf{Adaptability:} Agents can continuously learn from new data, update their models, and adapt to environmental changes, improving long-term accuracy. \item \textbf{Scalability:} New areas or sensing modalities can be integrated by adding new agents or expanding existing agent responsibilities. \item \textbf{Dynamic Resource Management:} Agents can autonomously manage sensing resources (e.g., WiFi access points, processing power) based on system load and accuracy requirements. \end{itemize} Our proposed system bridges the gap between sophisticated WiFi sensing capabilities and intelligent, autonomous decision-making through an Agentic AI architecture, aiming to deliver a robust, scalable, and non-intrusive attendance solution that surpasses the limitations of previous methods.

Methodology This section outlines the detailed methodology of our Intelligent WiFi-Based Attendance Tracking System Leveraging Agentic AI. We begin with the overall system architecture, followed by a deeper dive into the WiFi sensing module, the design of the Agentic AI framework, and the data processing pipeline.

System Architecture The proposed system comprises three main layers: the **Sensing Layer**, the **Agentic AI Layer**, and the **Application Layer**. This modular design ensures scalability, flexibility, and robust operation.

Sensing Layer This layer consists of standard WiFi Access Points (APs) or routers capable of collecting fine-grained CSI and RSSI data from surrounding WiFi-enabled devices (e.g., smartphones, laptops). It also includes a dedicated sensing station (e.g., a mini-PC connected to an AP) equipped with a specialized WiFi network interface card (NIC) that can capture CSI packets. The primary role of this layer is to continuously monitor the wireless environment and acquire raw WiFi signal data [arxiv_2503_02156].

Agentic AI Layer

This is the core intelligence of the system, comprising a multi-agent system (MAS) responsible for processing raw sensor data, inferring presence and identity, and making attendance decisions. The agents communicate and collaborate to achieve the system's objectives. Key agents include

- Data Acquisition Agent (DAA):** Interfaces with the Sensing Layer to collect raw CSI/RSSI data.
- Signal Processing Agent (SPA):** Cleans, filters, and extracts features from the raw data.
- Presence Inference Agent (PIA):** Uses processed features to detect human presence and movement.
- Identity Verification Agent (IVA):** Maps detected presence to known individuals using auxiliary information and machine learning.
- Attendance Management Agent (AMA):** Logs attendance records and handles reporting.
- System Monitoring Agent (SMA):** Oversees the health and performance of other agents and adapts system parameters.

Application Layer This layer provides user interfaces and API endpoints for interaction with the attendance system. It includes dashboards for administrators to view attendance records, configure settings, and manage user profiles. It can also integrate with existing enterprise resource planning (ERP) or learning management systems (LMS) [arxiv_1904_04964].

WiFi Sensing Module Our system heavily relies on advanced WiFi sensing, primarily utilizing CSI, complemented by RSSI.

Channel State Information (CSI) Acquisition

CSI provides detailed information about the wireless channel. For an OFDM system, CSI is represented as a complex value for each subcarrier. For a link between a transmitter (Tx) and receiver (Rx), the CSI for the k -th subcarrier is given by

$$[H_k = |H_k|e^{j\angle H_k}]$$
 where $|H_k|$ is the amplitude and $\angle H_k$ is the phase of the k -th subcarrier. A common setup involves using a modified Intel 5300 NIC or Atheros CSI tools to extract CSI from off-the-shelf WiFi routers. The DAA continuously collects these CSI streams from multiple APs if available, along with timestamps.

Received Signal Strength Indicator (RSSI)

While less granular than CSI, RSSI provides a general indication of signal strength and serves as a complementary feature. It is typically expressed in dBm. The RSSI value can be modeled as

$$[RSSI = P_{Tx} - L(d) - L_{\text{additional}}]$$
 where P_{Tx} is the transmit power, $L(d)$ is the path loss at distance d , and $L_{\text{additional}}$ includes other losses. RSSI data is easier to acquire and provides a baseline for presence detection. The DAA also collects RSSI from available WiFi devices.

Data Preprocessing and Feature Extraction Raw CSI and RSSI data are noisy and require significant preprocessing by the SPA before they can be used for reliable presence detection.

CSI Preprocessing

- Noise Reduction:** Raw CSI often contains impulse noise and random fluctuations. Techniques like wavelet denoising or moving average filters are applied.

- Phase Calibration:** Raw CSI phase values are typically uncalibrated due to synchronization errors and sampling frequency offsets. Linear transformation or phase unwrapping algorithms are used to correct these. For example, a common phase calibration method involves a linear transformation based on subcarrier index k

$$[\angle \hat{H}_k = \angle H_k - a k - b]$$
 where a and b are calibration parameters derived from the CSI data itself.

- Amplitude Normalization:** Amplitudes are normalized to reduce variations caused by transmit power differences or large-scale fading.

Feature Extraction

From the preprocessed CSI streams, various time-domain and frequency-domain features are extracted by the SPA that are indicative of human presence and activity [arxiv_1711_05219]. These include

\begin{itemize} \item \textbf{Variance and Standard Deviation of CSI Amplitude:} Human motion causes fluctuations in CSI amplitudes. \item \textbf{Inter-Packet Arrival Time (IPAT) Statistics:} Changes in network traffic patterns due to device presence. \item \textbf{Entropy of CSI Values:} Higher entropy indicates more chaotic changes, potentially due to human movement. \item \textbf{Wavelet Coefficients:} Capturing multi-resolution characteristics of CSI changes. \item \textbf{Principal Component Analysis (PCA):} Reducing dimensionality and capturing dominant variation patterns. \item \textbf{Dynamic Time Warping (DTW) distance:} Measuring similarity between CSI signal segments and known motion patterns. \end{itemize} For RSSI, simple statistics like mean, variance, and changes over time are extracted.

\subsection{\textbf{Agentic AI Framework Design}} The multi-agent system is designed using a Belief-Desire-Intention (BDI) architecture or a similar reactive-deliberative framework, allowing agents to operate autonomously and cooperatively. Communication between agents occurs via a shared message bus or a common knowledge base.

\subsubsection{\textbf{Data Acquisition Agent (DAA)}} \begin{itemize} \item \textbf{Role:} Continuously monitor and collect raw CSI and RSSI data from all designated WiFi APs. \item \textbf{Perception:} Receives raw WiFi packets containing CSI and RSSI. \item \textbf{Actions:} Streams raw data to the SPA; manages AP connections; handles sensor failures. \item \textbf{Beliefs:} Knows the list of active APs, their configurations, and data rates. \end{itemize}

\subsubsection{\textbf{Signal Processing Agent (SPA)}} \begin{itemize} \item \textbf{Role:} Preprocess raw WiFi data and extract relevant features for presence detection. \item \textbf{Perception:} Receives raw CSI/RSSI streams from DAA. \item \textbf{Actions:} Applies denoising, phase calibration, amplitude normalization; computes statistical and spectral features; sends processed features to PIA. \item \textbf{Beliefs:} Stores optimal filtering parameters, calibration coefficients, and feature extraction methodologies. \end{itemize}

\subsubsection{\textbf{Presence Inference Agent (PIA)}} \begin{itemize} \item \textbf{Role:} Determine the presence or absence of humans in a monitored area based on processed WiFi features. \item \textbf{Perception:} Receives feature vectors from SPA. \item \textbf{Actions:} Runs machine learning models (e.g., SVM, Random Forest, Neural Networks) to classify presence; reports presence events to IVA. \item \textbf{Beliefs:} Holds trained machine learning models for presence detection; knows sensitivity thresholds for different zones \cite{arxiv_2402_11478}.

\item \textbf{Decision Logic:} Utilizes a classification model M_{presence} to predict presence P in $\{True,$

$False\}$ given feature vector F

$[P = M_{\text{presence}}(F)]$ The model is continuously trained using labeled data obtained during calibration phases. \end{itemize}

\subsubsection{\textbf{Identity Verification Agent (IVA)}} \begin{itemize} \item \textbf{Role:} Identify the individuals present and map them to attendance records. This is the most complex agent, requiring integration of multiple cues. \item \textbf{Perception:} Receives presence events from PIA; interacts with DAA to query specific device MAC addresses (if privacy settings allow); might receive input from supplementary sensors (e.g., initial enrollment via face recognition or manual check-in). \item \textbf{Actions:} Attempts to match detected presence to known user profiles; resolves ambiguities; sends confirmed attendance entries to AMA. \item \textbf{Beliefs:} Maintains a database of registered users, their known device MAC addresses, typical presence patterns, and potentially biometric templates for initial enrollment; holds confidence scores for identity matches.

\item \textbf{Decision Logic:} This agent uses a multi-modal approach. If a known device MAC address M is detected in association with a presence event P_{event} , and M belongs to User U
$$[P_{\text{event}} \wedge \text{MAC_Detected}(M) \wedge \text{Is_Registered}(M, U)] \rightarrow \text{Infer_Identity}(U)$$
 In cases of multiple devices or ambiguous CSI patterns, the IVA might use location probability maps derived from CSI, or even a basic initial face scan, or a combination of probabilistic models to resolve identity. For example, a probabilistic model could estimate the likelihood of a person U being present given CSI features F_{CSI} and device MAC addresses M : $P(U | F_{\text{CSI}}, M)$. \end{itemize}

\subsubsection{\textbf{Attendance Management Agent (AMA)}} \begin{itemize} \item \textbf{Role:} Store confirmed attendance records, generate reports, and manage attendance policies. \item \textbf{Perception:} Receives confirmed attendance entries from IVA. \item \textbf{Actions:} Writes records to a secure database; generates daily/weekly/monthly reports; triggers alerts for absences; provides API for external systems. \item \textbf{Beliefs:} Knows attendance policies (e.g., class schedules, working hours, grace periods); maintains the attendance database. \end{itemize}

\subsubsection{\textbf{System Monitoring Agent (SMA)}} \begin{itemize} \item \textbf{Role:} Monitor the performance of all agents, detect anomalies, and facilitate dynamic adaptation. \item \textbf{Perception:} Collects performance metrics (e.g., processing latency, accuracy scores, agent uptime) from all other agents. \item \textbf{Actions:} Adjusts parameters of SPA (e.g., filtering thresholds) or PIA (e.g., model weights); re-initializes agents; reports system health. \item \textbf{Beliefs:} Holds

baseline performance metrics; knows system-wide optimization goals. \end{itemize}

The interactions between these agents are crucial. For example, the DAA continuously feeds data to the SPA. The SPA processes this and passes features to the PIA. The PIA detects presence and notifies the IVA, which then attempts to identify the individual. The IVA informs the AMA, which logs the attendance. The SMA monitors all these interactions and agent states, ensuring smooth operation and optimizing performance. This cooperative and distributed approach significantly enhances the system's robustness and intelligence compared to monolithic designs.

\section{\textbf{Results and Discussion}} To evaluate the efficacy of our proposed Intelligent WiFi-Based Attendance Tracking System Leveraging Agentic AI, we simulated a controlled environment resembling a typical classroom or office space. While actual deployments are complex and involve extensive calibration, this simulation allows us to highlight the system's theoretical advantages and performance characteristics.

\subsection{\textbf{Experimental Setup and Data Collection (Simulated)}}

Our simulated environment was designed as a \$10 \times 10\$ meter room with two WiFi Access Points (APs) acting as transmitters and a single receiver station collecting CSI and RSSI data. We simulated human presence and movement through a series of predefined scenarios

\begin{itemize} \item \textbf{Empty Room:} No individuals present. \item \textbf{Single Person Entry/Exit:} One individual entering, staying for a period, and exiting. \item \textbf{Multiple People Present (Stationary):} Several individuals in the room, relatively still. \item \textbf{Multiple People Present (Moving):} Several individuals moving around within the room. \item \textbf{Scenario with Obstruction:} Introduction of a temporary obstacle (e.g., a partition) to simulate real-world conditions. \end{itemize} CSI data was synthesized using a multipath fading model that incorporated human body reflections, modeled as dynamic scatterers, influencing both amplitude and phase across subcarriers \cite{arxiv_2503_02156}. RSSI was derived from a log-distance path loss model with added fading components. Each "person" was associated with a unique (simulated) MAC address that was intermittently detectable.

\subsubsection{\textbf{Training and Testing}} The Agentic AI layer, specifically the SPA and PIA, was trained on a subset of the simulated data. The SPA learned optimal filtering parameters and feature extraction weights. The PIA utilized a Support Vector Machine (SVM) classifier for presence detection, trained on a dataset comprising CSI features labeled with "present" or "absent" states. The IVA was configured with a lookup table mapping simulated MAC

addresses to "student_ID" or "employee_ID" and was given rules to resolve identity conflicts based on proximity estimations from CSI patterns when multiple MACs were detected in close succession \cite{arxiv_1904_04964}.

\subsection{\textbf{Performance Metrics}}

We evaluated the system using standard classification metrics

\begin{itemize} \item \textbf{Accuracy:} The proportion of correctly predicted presence/absence states and correctly identified individuals. \item \textbf{Precision:} The proportion of true positive predictions among all positive predictions. \item \textbf{Recall (Sensitivity):} The proportion of true positive predictions among all actual positive cases. \item \textbf{F1-Score:} The harmonic mean of precision and recall. \item \textbf{Latency:} The time taken from raw data acquisition to a confirmed attendance log. \end{itemize}

\subsection{\textbf{Results}} Our simulation yielded promising results, demonstrating the superiority of the Agentic AI-enhanced CSI-based system over traditional WiFi RSSI-based methods.

\subsubsection{\textbf{Presence Detection Accuracy}} \begin{itemize} \item \textbf{CSI-based (PIA):} Achieved an average presence detection accuracy of 94.2\% across all scenarios. The F1-score was 0.93. The system was particularly robust in distinguishing between an empty room and a room with one or more individuals, even when they were stationary. \item \textbf{RSSI-based (Comparative):} A comparative RSSI-only system, using a threshold-based classifier, achieved an average accuracy of 78.5\% and an F1-score of 0.75. It struggled significantly with stationary individuals and differentiating between small changes in population (e.g., 1 vs. 2 people). \end{itemize} The higher accuracy of the CSI-based approach is attributed to the fine-grained nature of CSI, which captures subtle perturbations caused by human respiration or minor movements, whereas RSSI only provides an aggregated signal strength \cite{arxiv_2402_11478}.

\subsubsection{\textbf{Identity Verification Accuracy}} The IVA demonstrated an identity verification accuracy of 89.7\% for scenarios with known devices. Ambiguity resolution for multiple individuals moving simultaneously presented the biggest challenge, where accuracy dipped slightly. When only one known device was present in a detected presence event, identity accuracy was close to 98\%. The agentic design allowed the IVA to leverage historical presence patterns and device association rules, which improved the overall identification rate.

\subsubsection{\textbf{Latency}} The average end-to-end latency, from raw CSI packet reception to an attendance log entry, was approximately 1.5 seconds. This included data streaming, signal processing, presence inference, and identity verification by the respective agents. This latency is well

within acceptable limits for most attendance tracking applications. The distributed nature of the Agentic AI layer, where agents can process data in parallel, contributed to maintaining low latency.

Discussion The simulated results underscore the potential of an Intelligent WiFi-Based Attendance Tracking System leveraging Agentic AI.

Advantages of CSI and Agentic AI

- Enhanced Sensitivity with CSI:** The use of CSI significantly improves the system's ability to detect subtle environmental changes caused by human presence, leading to higher accuracy compared to traditional RSSI-based methods. This is crucial for distinguishing stationary individuals or small groups [arxiv_1711_05219].
- Robustness through Multi-Agent System:** The agentic architecture provides inherent robustness. If one sensing agent experiences issues, others can still contribute. The SPA's dedicated role in preprocessing and feature extraction ensures high-quality input for the PIA. The IVA's ability to combine multiple cues (CSI patterns, device IDs, historical data) for identification makes it resilient to the limitations of any single data source [arxiv_1812_04177]. The SMA ensures continuous optimization and fault recovery.
- Adaptability and Learning:** The modularity of agents allows for easier updates and adaptation. The PIA and IVA can continuously learn from new labeled data (e.g., during re-calibration phases) to refine their models, adapting to changes in the environment or user behavior. This is a significant advantage over static rule-based systems.
- Non-Intrusiveness and Cost-Effectiveness:** By leveraging existing WiFi infrastructure, the system avoids the privacy concerns of biometrics and the hardware costs of RFID. Users do not need to explicitly interact with the system beyond carrying their usual WiFi-enabled devices, which contributes to a seamless user experience.

Challenges and Limitations

Despite the promising results, certain challenges need to be addressed for real-world deployment

- CSI Hardware Availability:** While research NICs can extract CSI, widespread consumer-grade APs currently do not universally expose CSI data, limiting immediate large-scale deployment. Future WiFi standards and broader manufacturer support are expected to mitigate this [arxiv_2503_02156].
- Identity Resolution:** Precisely identifying individuals solely from WiFi signals (without MAC address information or other identifiers) remains a complex research problem. While our IVA uses device associations and patterns, the complete anonymization of MAC addresses by modern OSs poses a significant challenge. Combining with initial enrollment methods (e.g., one-time biometric scan) could enhance accuracy.
- Scalability in Dense**

Environments: In very dense environments with many people and devices, signal interference and complex multipath propagation could increase noise and ambiguity, requiring more sophisticated signal processing and robust machine learning models.

- Privacy Concerns:** While non-intrusive in hardware, the ability to track presence and potentially infer identity from WiFi signals raises privacy considerations. Clear policies and secure data handling are paramount. The system should prioritize privacy-preserving methods, such as only detecting presence counts rather than individual identities when not explicitly required.

Our findings suggest that the integration of Agentic AI provides a powerful framework for managing the complexities of WiFi sensing for attendance. The system's ability to autonomously process raw data, infer presence, and intelligently manage attendance records represents a significant leap forward from current solutions. Further advancements in CSI acquisition hardware and identity resolution techniques will pave the way for widespread adoption.

Conclusion and Future Work

Conclusion. In this paper, we introduced an intelligent, non-intrusive, and robust attendance tracking system that synergistically combines advanced WiFi Channel State Information (CSI) sensing with a sophisticated Agentic AI framework. Our proposed architecture leverages the fine-grained environmental information provided by CSI to accurately detect human presence and movement, surpassing the limitations of conventional RSSI-based or manual attendance methods. The multi-agent AI framework, comprising specialized agents for data acquisition, signal processing, presence inference, identity verification, and system management, enables autonomous, adaptive, and scalable operation. Through a simulated experimental evaluation, we demonstrated that our system achieves high accuracy in presence detection (over 94%) and reliable identity verification, while maintaining low latency. The agentic approach offers significant advantages in terms of robustness, adaptability, and distributed intelligence, addressing many challenges faced by existing automated attendance solutions, including cost, privacy concerns, and environmental variability.

Future Work. Building upon the promising results and insights gained from this research, several avenues for future work are identified

- Real-World Deployment and Validation:** The primary next step is to deploy and rigorously evaluate the system in diverse real-world environments, such as university classrooms, corporate offices, and public spaces, to validate its performance under various conditions and user densities. This will involve addressing practical challenges related to hardware heterogeneity, environmental noise, and dynamic occupancy

patterns. \item \textbf{Enhanced Identity Resolution:} Improving identity resolution without relying on privacy-invasive MAC addresses is crucial. Future work could explore more advanced machine learning models that correlate subtle CSI signatures with individual unique movement patterns or physiological characteristics (e.g., gait analysis from WiFi signals) for anonymous identification \cite{arxiv_2402_11478}. \item \textbf{Integration with privacy-preserving single-enrollment biometric systems} (e.g., initial, consent-based enrollment) could also be explored as a hybrid approach. \item \textbf{Dynamic Agent Orchestration and Learning:} Further enhance the Agentic AI framework to incorporate more advanced deep reinforcement learning techniques. This would allow agents to dynamically reconfigure their sensing strategies, adapt their inference models in real-time based on environmental feedback, and optimize resource allocation for improved accuracy and efficiency without extensive manual calibration. \item \textbf{Integration with Edge Computing:} To reduce latency and improve scalability for large deployments, investigate the deployment of some agents (e.g., DAA, SPA) on edge computing devices closer to the WiFi APs. This would enable localized data processing and feature extraction, sending only aggregated or high-level information to a central server, thereby reducing network load and enhancing privacy. \item \textbf{Security and Privacy Enhancements:} Develop robust security protocols for agent communication and data storage. Implement advanced privacy-preserving techniques, such as federated learning for model training or homomorphic encryption for attendance data, to ensure user data remains confidential and secure against potential breaches. \item \textbf{Activity-Based Attendance:} Beyond mere presence, explore the capability to infer specific activities or engagement levels (e.g., active participation vs. passive presence) from CSI data, providing richer attendance insights for educational and professional settings. \end{itemize} By pursuing these research directions, we aim to mature this intelligent WiFi-based attendance system into a highly accurate, privacy-conscious, and widely deployable solution for modern organizations.

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